



Invention Triage Assessment | UAIPD 26-0030

Cooperative Transmission Authority Control for Congestion-Resilient Basic Safety Message Dissemination over C-V2X PC5 Sidelink

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Overview

Cooperative Transmission Authority Control is a concept-stage (early-stage algorithm design; no prototype and no measured performance) software/middleware layer that operates above the C-V2X PC5 radio-resource-selection layer and coordinates which vehicle On-Board Units (OBUs) may transmit routine Basic Safety Messages (BSMs) in a given time epoch. It organizes nearby OBUs into dynamic "cohorts," grants full-BSM transmission authority to active cohorts on a time-indexed schedule while inactive cohorts defer, suppress, or compress, and layers on an absolute safety override, age-of-information (AoI) limits, and a legacy fallback. The commercial thesis rests on a durable, regulation-driven need: as C-V2X penetration rises, uncoordinated BSM broadcasting can congest the limited PC5 sidelink in dense traffic, and a standard-agnostic software overlay could improve packet delivery without new radio hardware. Its central, unresolved tensions are three: (1) the problem is largely anticipatory — meaningful PC5 congestion emerges mainly at high penetration expected in the 2030s; (2) the concept sits in an exceptionally dense, standardized, and SEP-heavy field, where cluster/TDMA VANET MAC, cooperative/platoon C-V2X resource allocation, and 3GPP Release-17 inter-UE coordination all teach closely related ideas; and (3) no empirical evidence yet supports the performance or non-obviousness claims. The disposition below is contingent on a credible simulation study and a focused freedom-to-operate (FTO)/SEP read.

Opportunities	Risks
• Durable regulatory tailwind: The FCC reserved 30 MHz of the 5.9 GHz band for C-V2X, USDOT issued its Aug-2024 "Saving Lives with Connectivity" deployment plan, and NHTSA's 2024 V2X NPRM proposes light-vehicle V2X capability (phased ~2027–2029) — all pointing to the rising penetration that is the precondition for the congestion this invention targets. • Capital-light, licensable model: Architected as software/firmware/middleware above the PC5 resource-selection layer, it can deploy via OTA/stack updates with no new radio hardware, fitting OEM, Tier-1, chipset, or RSU licensing and co-development. • Clean, undisclosed IP (pending verification): The disclosure reports internal development, no external/sponsored funding, and no prior public disclosure — preserving broad filing options if the absence of a federal-funding nexus is confirmed. • Genuine (if narrow) white space: Implementing cooperative "transmission-authority" scheduling strictly as a standard-agnostic overlay above the MAC/PHY — rather than inside the radio —	• Anticipatory market: At ~15–20% new-vehicle C-V2X penetration in 2026, real PC5 congestion is rare; the pain point materializes mainly at high penetration (2030s), pushing meaningful revenue years out. • Very dense, standardized prior art: Cluster/TDMA VANET MAC, cooperative/platoon C-V2X resource allocation, AoI-aware C-V2X scheduling, and 3GPP Rel-17 inter-UE coordination all teach closely related grouping/turn-taking/coordination — undercutting the 95% automated novelty score. • SEP / FTO exposure: The field is dominated by telecom SEP holders (Qualcomm, LG, Huawei, Ericsson, Nokia, InterDigital, Ofinno); broad claims face high freedom-to-operate risk, and a formal SEP/FTO read is warranted before spend. • No empirical evidence: Concept-stage with no simulation or measured metrics (packet-delivery ratio, channel busy ratio, latency, AoI vs. baseline SPS+DCC) — the single most decision-relevant gap for both non-obviousness and commercial claims.



Opportunities	Risks
is comparatively less crowded than resource-level solutions. - Multiple commercialization channels: Chipset/stack vendors (Qualcomm/Autotalks, Commsignia, Cohda Wireless), RSU/ITS integrators (Kapsch, Iteris, Derq, Econolite), Tier-1s (Bosch, Continental, Denso, Aptiv, Harman), and public-sector V2X pilots. - Infrastructure-assisted near-term angle: RSU/edge-computed cohort scheduling aligns with the signalized-intersection deployments IIJA-funded ITS programs are prioritizing, offering a pilot path ahead of vehicle-fleet penetration.	- Adoption friction: Adding a non-standardized coordination layer to functional-safety-governed (ISO 26262) V2X stacks — with GNSS-timing, control-overhead, hidden-node, and legacy-mixing concerns — is a high bar for OEM/Tier-1 acceptance. - Standardization risk: A mechanism that duplicates or competes with standardized DCC and inter-UE coordination may see limited proprietary-licensing leverage unless contributed to, or made complementary with, 3GPP/SAE/ETSI/5GAA standards.

Executive Summary

C-V2X PC5 sidelink enables direct vehicle-to-vehicle safety messaging without cellular infrastructure. In dense traffic, many OBUs contend for limited PC5 resources at once, raising channel busy ratio, packet collisions, and message latency. Today's standardized congestion control — ETSI Decentralized Congestion Control and 3GPP channel-occupancy limits, layered over sensing-based Semi-Persistent Scheduling (SPS) — acts independently at each vehicle by adjusting transmission rate, power, packet size, or reservation interval. The invention proposes a cooperative layer that instead coordinates which vehicles transmit in each epoch, aiming to prevent the simultaneous access attempts those per-vehicle methods cannot.

The opportunity is real but narrower than the disclosure frames it. The strongest structural asset is a durable, mandate-driven tailwind (FCC 5.9 GHz allocation, USDOT deployment plan, NHTSA V2X rulemaking) paired with a capital-light, licensable software model. The weakest fit is timing: at current penetration the PC5 channel is rarely congested, so the value the invention creates is largely future-tense and crystallizes only as C-V2X becomes ubiquitous. The addressable revenue is a thin coordination/congestion-control slice of the C-V2X software segment — not the broad multi-billion-dollar "V2X market" that automated scoring implies.

From an IP standpoint, the core ideas — grouping vehicles into cohorts, time-indexed turn-taking, Aol-aware deferral, safety-priority override, and legacy fallback — are each well represented in the C-V2X and VANET literature (cooperative/platoon resource allocation, cluster-based TDMA MAC, Aol-aware scheduling) and in standardized 3GPP Release-17 inter-UE coordination. Genuine, but narrow, novelty remains at the specific integration: a standard-agnostic overlay that gates transmission authority above the resource-selection layer, plus the particular Aol-bounded deferral and fallback state machine. Any filing should be scoped tightly to those sub-components; broad claims on "grouping and scheduling V2X transmissions" are unlikely to survive examination and carry substantial freedom-to-operate risk in a SEP-dense field.

Overall, this is a borderline, higher-tailwind but crowded and pre-empirical opportunity that merits a low-cost, tightly gated path rather than immediate heavy investment. The recommended posture is to withhold significant spend until a credible network-simulation study establishes measurable gains over baseline SPS+DCC and Release-17 inter-UE coordination; order a focused FTO/SEP read; and, if the data support it, file a narrowly scoped provisional while engaging 3GPP/SAE/5GAA standards as a complementary-positioning and defensive hedge. Because the prior-art field is deep and fast-moving, a low-cost provisional timed to the first credible simulation evidence is the pragmatic play.

Value Proposition



The invention proposes an intermediate "coordination tier" between two existing extremes: purely decentralized per-vehicle congestion control (which does not coordinate who transmits) and hardware-based capacity upgrades (e.g., migrating to wider-bandwidth 5G NR-V2X). As a software overlay, it promises higher BSM delivery reliability under congestion — fewer collisions, lower latency, bounded information age — deployable via OTA/firmware updates without new radios. Where the channel is genuinely congested, the economic logic is straightforward: more reliable safety messaging at the moments density is highest, at negligible incremental hardware cost.

The value proposition should be stated with two caveats attached. First, it is contingent on penetration reaching congestion levels; below that threshold the overlay adds coordination overhead without a congestion problem to solve, and control signaling can itself consume scarce channel capacity. Second, the claimed advantage is relative to standardized SPS+DCC and inter-UE coordination, so the value depends on the overlay demonstrably outperforming those baselines in realistic, high-density simulation — evidence that does not yet exist. The office should frame the technology as a standard-agnostic, infrastructure-capable enhancement (and a candidate standards contribution), not as a replacement for standardized mechanisms, to keep both the FTO posture and the adoption story credible.



Technology Overview

The Problem

In high-density traffic environments, such as urban intersections and freeway bottlenecks, the simultaneous broadcast of Basic Safety Messages (BSMs) by numerous vehicles over limited Cellular Vehicle-to-Everything (C-V2X) PC5 sidelink resources creates severe wireless channel congestion. This uncoordinated mass access leads to high channel busy ratios, resource contention, packet collisions, and increased message latency, which can critically undermine vehicular safety applications. While existing congestion-control methods attempt to mitigate this issue by having individual vehicles independently adjust parameters like transmission rate, power, packet size, or resource reservation intervals, these decentralized approaches operate in silos. Because they lack a mechanism to coordinate transmission timing among neighboring vehicles, current methods ultimately fail to prevent simultaneous access attempts and subsequent packet loss under extreme vehicle densities.

The Solution

The invention introduces a cooperative, software-based scheduling layer operating above the C-V2X PC5 radio-resource selection layer to mitigate wireless channel congestion during Basic Safety Message (BSM) dissemination. Technically, the system dynamically organizes vehicle On-Board Units (OBUs) into local communication cohorts and employs a time-indexed epoch schedule, granting full BSM transmission authority exclusively to active cohorts while inactive cohorts defer or suppress their broadcasts. To ensure reliability, the protocol integrates an absolute safety override for immediate emergency transmissions, age-of-information constraints to maintain data freshness, and fallback mechanisms to support legacy, non-cooperative vehicles. This approach is highly novel because, whereas existing congestion-control methods operate independently at the individual OBU level by merely adjusting parameters like transmission rate or power, this invention cooperatively coordinates exactly *which* vehicles are authorized to transmit at any given time, effectively preventing simultaneous access attempts and packet collisions in high-density traffic environments.

Technology Readiness Level: TRL 2 (Concept / Early-Stage Algorithm Design)

Per the disclosure, the protocol has been conceptually reasoned but no functional prototype exists and no empirical or simulation performance data have been generated. At this stage, essentially all differentiating claims — congestion reduction, latency and Aol improvement, overhead tolerability, timing robustness, and graceful legacy coexistence — are unverified. The upside of the early stage is that the university holds an unencumbered, undisclosed position to capture foundational IP if the mechanism validates and proves non-obvious over the dense baseline art.

Development Path

Near-term milestones required to reach a defensible position and a data-supported filing:

1. Build a network-simulation testbed (e.g., ns-3 / OpenCV2X or Artery+Veins+SUMO) and quantify packet-delivery ratio, channel busy ratio, latency, and Aol versus baseline SPS+DCC and versus Release-17 inter-UE coordination across a sweep of vehicle densities and penetration rates — the single most decision-relevant data point.
2. Specify and characterize the cohort-formation and authority-granting algorithm, and measure its control-signaling overhead to confirm it does not paradoxically worsen congestion.
3. Analyze robustness to GNSS/timing drift (urban canyons, tunnels), hidden-node and cohort-boundary collisions, mixed legacy traffic estimation, and safety-override latency.
4. Define the fallback state machine and the Aol-bounded deferral rules precisely enough to support narrow claims.



5. Map the mechanism against standardized DCC and inter-UE coordination to isolate the defensible novelty and assess standards-contribution potential.

Patentability Assessment

Novelty

The individual building blocks are all established art: grouping/clustering vehicles, TDMA-style time-slotted turn-taking, cooperative resource allocation, Aol-aware scheduling, QoS/priority preemption, and legacy fallback each have dense prior art. More importantly, the specific combination is substantially anticipated in the C-V2X-specific literature (cooperative and platoon resource-allocation protocols, cluster-based TDMA MAC, and Aol-aware platoon C-V2X scheduling) and in standardized 3GPP Release-17 inter-UE coordination, which already lets one UE coordinate the resources another UE uses. Residual, genuinely novel white space is narrow but real and sits at the integrated overlay rather than the concept: (i) a standard-agnostic layer that gates transmission authority above resource selection (as opposed to modifying the MAC/PHY or the resource pool); (ii) the specific Aol-bounded deferral/compression logic tied to cohort state; and (iii) the particular degrade-and-fallback state machine for mixed legacy environments. Patentability depends on claiming these sub-components, not the general idea.

Non-Obviousness

Combining vehicle grouping, turn-taking authority, DCC-style triggers, Aol limits, and a safety override is likely to be characterized by an examiner as obvious-to-combine given the density of each element in the art. Non-obviousness therefore cannot rest on "coordinating which vehicles transmit for V2X." It can more plausibly rest on a specific, non-trivial authority-granting mechanism that measurably outperforms standardized SPS+DCC and inter-UE coordination under realistic density — an argument the disclosure does not yet support because no simulation or measured evidence exists. A strong defense will require empirical results demonstrating a real, repeatable performance delta over the standardized baselines.

Utility

Utility is strong and clearly articulated: congestion-resilient BSM dissemination for dense traffic, with direct application to connected-vehicle safety, intelligent transportation systems, autonomous fleets, and public smart infrastructure. Utility is not the constraint here; novelty/non-obviousness and freedom-to-operate are.

Prior Art Summary

The decentralized V2X congestion-control and sidelink resource-selection landscape is dense and accelerating, spanning academic protocols and a deep telecommunications patent estate. Broad claims are at high risk; freedom-to-operate depends on integrated-system specificity. Representative references to review with counsel (illustrative, not a professional search):

- Cooperative / platoon resource-allocation for C-V2X Mode 4 (e.g., "A Cooperative Resource Allocation Protocol for Platoon Communication in C-V2X," 2026; "Context-aware / Cooperative Resource Allocation for V2V in C-V2X," 2024) — coordinated grouping and scheduling over SPS.
- Aol-aware, group-based C-V2X scheduling (e.g., "Aol-Aware Resource Allocation for Platoon-Based C-V2X Networks via Multi-Agent Reinforcement Learning," 2021; "Age of Information Optimized MAC in V2X Sidelink," 2020) — closest art to the age-of-information trigger.
- Cluster-based TDMA MAC for VANETs (e.g., TC-MAC; adaptive/decentralized TDMA BSM frameworks; scalable TDMA cluster MAC) — decades of art on cluster-head-coordinated turn-taking transmission.
- 3GPP Release-16/17 sidelink inter-UE coordination — e.g., US 11,690,048 (inter-UE coordination signaling); WO2022240662A1 and WO2022147228A1 (resource selection for inter-UE coordination); US 2021/0321430 and US 2021/0258114 (V2X inter-UE coordination) — standardized coordination of which resources a UE uses.



- Foundational C-V2X SPS/DCC scheduling art from telecom SEP holders (Qualcomm and others) governing sensing-based semi-persistent scheduling and congestion parameters on the PC5 sidelink.

Freedom-to-Operate (Initial Scan)

FTO risk is elevated because the field is both academically saturated and dominated by standard-essential patents. Broad claims on generic "group + schedule V2X transmissions" or "AoI-triggered V2X updates" are high-risk. A cleaner path exists if claims are scoped tightly to the standard-agnostic transmission-authority overlay, the specific AoI-bounded deferral/compression logic, and the degrade-and-fallback state machine — explicitly positioned above, and complementary to, standardized DCC and inter-UE coordination. This is a formal FTO/SEP-opinion candidate before any spend beyond a provisional; the Release-17 inter-UE-coordination filings and the telecom SEP estate in particular warrant a focused read.



Market Analysis & Competitive Landscape

Regulatory & Standards Context (current as of July 2026)

The regulatory backdrop is a net positive but penetration- and rule-gated. In 2020 the FCC reallocated the 5.9 GHz band and reserved 30 MHz (5.895–5.925 GHz) for C-V2X, effectively ending the DSRC-vs-C-V2X standards contest in the U.S.; however, the FCC's Second Report and Order with final operating rules has remained pending (interim waivers were granted starting 2023), which has slowed some deployment commitments. NHTSA's 2024 V2X Notice of Proposed Rulemaking proposes requiring V2X capability in new light vehicles (phased compliance discussed for roughly 2027–2029), and USDOT's Aug-2024 "Saving Lives with Connectivity" plan plus roughly \$5B of IIJA-linked funding provide a deployment push. Standards continue to evolve (3GPP Release-16/17 NR-V2X and inter-UE coordination; SAE J2945/J3161; ETSI ITS; 5GAA profiles). Net: sustained, mandate-driven demand is likely, but the specific congestion problem this invention addresses is anchored to high penetration that is still several years out, and the final FCC rules and NHTSA mandate are not yet locked.

Target Market

Three verticals, ordered here by fit:

- **Vertical 1 — In-vehicle C-V2X stack / chipset licensing (OEM, Tier-1, chipset)**
 - The largest long-run addressable value, but the highest adoption bar: any coordination layer added to safety-critical, standardized stacks must clear functional-safety (ISO 26262), interoperability, and standards-alignment hurdles. Best-fit partners: Qualcomm (now including Autotalks), Commsignia, Cohda Wireless, Harman, Bosch, Continental, Denso, Aptiv.
- **Vertical 2 — Infrastructure-assisted / RSU-side coordination (state DOTs, ITS integrators)**
 - The most credible near-term pilot pathway: RSUs can compute and broadcast cohort schedules at signalized intersections and dense corridors, aligning with IIJA-funded ITS deployments. Best-fit partners: Kapsch TrafficCom, Iteris, Derq, Econolite, Applied Information, Miovision, Commsignia RSUs.
- **Vertical 3 — AV / robotaxi & shuttle fleet cooperative perception (geofenced dense operations)**
 - A real conceptual fit, but a small and consolidating buyer set. Note an accuracy correction to the automated report: GM shut down Cruise's robotaxi program (Dec 2024) and confirmed it will not relaunch, so Cruise is not a viable partner; Waymo, Zoox, May Mobility, Beep, and Baidu Apollo remain the relevant names.

Market Size & Trends

Credible third-party estimates place the U.S. C-V2X market at roughly \$1.2–1.6B in 2026 growing to about \$8.5–11.5B by 2035 (~18-28% CAGR), and the global market near \$1.7B in 2025 growing at ~33% CAGR; hardware (OBUs, RSUs, chipsets) dominates today (~55-60% of value), while software and security — the segment this invention would sit in — is ~12-15% of value now, rising to ~35–40% by 2035. The invention competes only for a thin coordination/congestion-control slice of that software segment. The automated report's per-use-case figures (\$20B freight/toll, \$3B smart-intersection, \$800M AV, with ~40% average growth) should be treated as loosely scoped and optimistic — directional context, not addressable revenue.

Competitive Advantages

- **Coordinated Congestion Control vs. Independent Decentralized Congestion Control:** The invention uses a cooperative scheduling layer to dynamically coordinate transmission authority among vehicles. This improves upon existing ETSI ITS-G5 Decentralized Congestion Control and standard C-V2X Mode 4



Sensing-Based Semi-Persistent Scheduling, which operate independently at the individual vehicle level and fail to prevent simultaneous access attempts in ultra-dense traffic scenarios.

- **Software-Only Deployment vs. Hardware Upgrades:** The invention operates as a software layer above the existing C-V2X PC5 medium access control layer. This provides a significant financial advantage over migrating to newer hardware-based scheduling solutions like 5G NR V2X, as it can be deployed via over-the-air software or firmware updates without requiring costly new radio hardware from automotive original equipment manufacturers.
- **Absolute Safety Override vs. Strict Time-Slotted Scheduling:** The invention incorporates an absolute safety override mechanism that allows vehicles to immediately bypass the scheduling protocol for critical events. This improves upon traditional Time Division Multiple Access MAC protocols and standard Semi-Persistent Scheduling, where emergency messages often experience extended delays while waiting for the next reserved time slot or reselection counter to expire.
- **Mixed-Environment Compatibility vs. Strict Proprietary Protocols:** The invention reserves channel capacity for legacy, non-cooperative vehicles and includes an automated fallback to standard uncoordinated autonomous resource selection. This improves upon strict proprietary cooperative scheduling protocols that degrade significantly when introduced into mixed-equipment environments containing legacy C-V2X Mode 4 vehicles.

Barriers to Commercialization

- **GNSS Dependency and Timing Synchronization:** Time-indexed epoch scheduling requires microsecond-level synchronization across all participating vehicles. In environments with degraded GNSS reception, such as urban canyons or tunnels, timing drift can cause scheduled epochs to overlap, resulting in packet collisions and protocol failure.
- **Control Signaling Overhead:** Dynamically organizing vehicles into cohorts and negotiating transmission schedules requires the continuous exchange of coordination messages. In highly dense traffic, the bandwidth consumed by this control signaling can paradoxically exacerbate channel congestion and increase overall network latency.
- **High Mobility and Topology Volatility:** Vehicular networks experience rapid topology changes due to high speeds and varying trajectories. Continuously updating local domains and cohort memberships in real-time is computationally intensive, and delays in state updates can lead to misallocated transmission authorities and dropped safety messages.
- **Boundary Interference and Hidden Nodes:** Because cohorts are formed based on localized spatial domains, vehicles at the edges of adjacent domains may not be aware of neighboring schedules. This creates a hidden node problem where simultaneous transmissions from active cohorts in overlapping boundary zones cause localized packet collisions.
- **Dynamic Legacy Traffic Estimation:** The system must dynamically partition PC5 channel resources to accommodate uncoordinated legacy vehicles. Accurately estimating the density and resource demands of these non-participating vehicles in real-time is highly complex, and miscalculations will lead to either wasted bandwidth or severe interference between scheduled and legacy transmissions.



Commercialization Pathways

Current Funding & IP Position

Per the disclosure, development has been internal, with no external or sponsored funding listed and no public disclosure to date — preserving the ability to file. Two items warrant verification before the office asserts a fully "clean" position. First, because the inventors are transportation researchers, confirm the absence of any federal grant nexus (USDOT/University Transportation Center, NSF, or similar) in the underlying work that would trigger Bayh-Dole reporting or government march-in rights. Second, confirm that the reported "background discussions with five to six companies over the past two years" were conducted under NDA and without enabling disclosure, so that novelty and filing scope are not impaired.

Grant Potential

The technology aligns well with the federal V2X-deployment agenda: USDOT ITS Joint Program Office, SMART grants, and V2X deployment programs; NSF Cyber-Physical Systems and SBIR/STTR; DOT University Transportation Centers; and state-DOT connected-vehicle pilots. Non-dilutive funding tied to an RSU-assisted, infrastructure-side demonstration is a low-cost way to generate the very simulation/field evidence the IP case needs while building a deployment narrative. Early engagement with 3GPP, SAE, and 5GAA standards groups is a further low-cost way to sharpen positioning and build optionality.

Strategic Partners

The most credible partners hold adjacent product lines into which a validated overlay could slot. These are licensing / co-development / field-validation and, in several cases, FTO targets — not confirmed interest:

- Chipset / V2X stack: Qualcomm (which now owns Autotalks — the automated report lists these as two separate partners), Commsignia, Cohda Wireless, Nordsys, Unex, Danlaw.
- RSU / ITS infrastructure: Kapsch TrafficCom, Iteris, Derq, Econolite, Applied Information, Miovision — the strongest near-term, infrastructure-assisted path.
- Tier-1 / OEM: Bosch, Continental, Denso, Aptiv, Harman; OEM programs at Ford, GM, Toyota, Hyundai, Volkswagen.
- Telecom / SEP holders: Qualcomm, Ericsson, Nokia — simultaneously potential licensees and freedom-to-operate considerations.
- Public sector: USDOT and state DOTs, via V2X deployment and smart-intersection programs.



Final Triage Recommendation

Primary Strength

A durable, regulation-driven tailwind (FCC allocation, USDOT plan, NHTSA rulemaking momentum) combined with a capital-light, licensable software model, a clean and undisclosed IP position, and a genuine — if narrow — white space at the standard-agnostic overlay abstraction. An infrastructure-assisted (RSU-side) path offers a plausible near-term pilot ahead of vehicle-fleet penetration.

Primary Risk

An anticipatory market whose value crystallizes years out at high penetration, intersecting an exceptionally dense, standardized, SEP-heavy prior-art field, with no empirical evidence and high adoption friction into functional-safety-governed stacks. Together these compress both the defensible novelty (to specific sub-components) and near-term revenue (to a thin software slice), against a mature standards ecosystem in which competing mechanisms are already standardized.

Suggested Next Steps

1. Gate on data: require a credible network-simulation study quantifying gains over baseline SPS+DCC and Release-17 inter-UE coordination across densities/penetrations before committing meaningful spend. Simulated performance is the single decision-driving metric.
2. Order a focused FTO/SEP read of the Release-17 inter-UE-coordination filings, the telecom SEP estate, and the VANET cluster-MAC / cooperative-C-V2X literature before drafting claims.
3. If the data support it, file a narrowly scoped provisional on the transmission-authority overlay, the AoI-bounded deferral/compression logic, and the fallback state machine — explicitly avoiding broad "group-and-schedule V2X" claims. Time the filing to the first credible simulation evidence given the fast-moving art.
4. Engage 3GPP / SAE / 5GAA standards to position the mechanism as complementary to (not duplicative of) standardized congestion control, and to build a standards-contribution/defensive-publication hedge.
5. Pursue USDOT/NSF non-dilutive funding and an RSU-assisted infrastructure pilot with a state DOT to generate evidence and a deployment narrative at low cost.
6. Verify the funding/disclosure cleanliness (no federal-grant nexus; NDA-covered company discussions) before relying on a broad filing posture.

Triage Recommendation: **Medium priority**

The invention merits a low-cost, tightly gated path rather than immediate significant investment. Its regulation-driven tailwind, clean and undisclosed IP, and capital-light licensable model are real assets; its dense standardized prior art, SEP/FTO exposure, absent empirical evidence, and anticipatory market are real constraints. Recommend revisiting upon delivery of a credible simulation study and a focused FTO/SEP read: a strong, repeatable performance delta over standardized baselines plus a clean FTO read would raise this rating and justify a narrow provisional, while confirmation of dense blocking art or negligible gains over standardized mechanisms would argue for a defensive-publication or standards-contribution posture instead.

Notice: This triage assessment was prepared for internal preliminary evaluation using the inventor disclosure, an AI-assisted IP/market analysis (Tradespace, TurbolInnovate), curated data exports, and current web research. Third-party market figures and algorithmic scores vary by source and are directional only. These findings are not formal expert advice or a legal patentability/FTO opinion; a registered patent attorney should conduct any formal novelty and freedom-to-operate analysis before filing.